Syllabus Year: 2024-2025

CHRIST (Deemed to be University), Bengaluru - 560029

End Semester Examinations - November 2025

Bachelor of Computer Applications SEMESTER - I

Course Code: BCA103-1 **Max Marks: 50**

Course Name: PRINCIPLES OF SOFTWARE DEVELOPMENT - I **Duration: 02:00**

General Instructions:

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / course title and number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer book. No sheets should be detached from the answer book.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer book.

Course Outcome: Students will be able to:

CO1: Design and Develop ethical software product.

CO2: Apply terms, concepts, and tools of relational database management systems.

CO3: Construct database projects using DDL and DML queries.

Section A **2*5=10**

1)	Analyze how views in a relational database system can be used to provide different perspectives on data without altering the underlying tables. Differentiate between a simple view and a complex view with examples.	**2**	(CO2, L4)
2)	Analyze how modularity in software development enhances the maintainability and flexibility of an ethical software product.	**2**	(CO1, L4)
3)	Examine how ethical considerations can influence the requirement elicitation process and affect the quality of the final software product. Provide one example to illustrate your answer.	**2**	(CO1, L4)
4)	Analyze how DDL and DML commands can be used to enforce data integrity in a relational database. Provide examples of each command type and explain their role.	**2**	(CO3, L4)
5)	Examine how normal forms can be used to differentiate between different layers of a relational database system.	**2**	(CO2, L4)

Section B **5*4=20**

6)	Construct a database schema using DDL commands for a library management system that includes tables for Books, Authors, Publishers, and Borrowers. Additionally, demonstrate how views can simplify querying operations by creating a view that shows all books borrowed by a specific borrower. Explain your design choices and the benefits of using this approach.	**5**	(CO3, L3)
7)	Describe the process of identifying and documenting software requirements using interviews and walkthroughs as elicitation techniques, and illustrate your explanation with an example scenario.	**5**	(CO2, L2)
8) a)	Demonstrate how the concept of normalization can be applied to avoid data redundancy and improve integrity in a relational database management system by constructing an appropriate example schema.	**5**	(CO2, L3)

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OR

b)	Determine the potential quality issues that can arise during database design phase and explain how to mitigate these issues using appropriate views.	**5**	(CO2, L3)
9)	Analyze how the use of UML diagrams during requirement elicitation can enhance the quality of software development.	**5**	(CO2, L4)

Section C **10*2=20**

10)	Examine how design patterns can be effectively utilized to address common software development issues during the requirement analysis phase. Provide specific examples of design patterns that are particularly useful for enhancing system scalability, maintainability, and security.	**10**	(CO1, L3)
11) a)	Discuss the significance of understanding different layers in the development process and explain how each layer contributes to ethical software development. Additionally, identify any potential ethical issues that might arise at each layer.	**10**	(CO1, L2)

OR

b)	Explain the concept of an entity view and discuss its importance in database design. Further, illustrate how to create, insert data into, and delete a view related to entities in a university management system. Provide examples to support your explanation.	**10**	(CO1, L2)

Revised Bloom's Taxonomy (RBT) Levels

L1 - Remember
L2 - Understand
L3 - Apply
L4 - Analyse
L5 - Evaluate
L6 - Create

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